

LAB EXPERIMENT RECORD

03-10-2026---Set 9: Online Learning Activity Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	03 OCTOBER 2026
Day	SATURDAY

Aim

To analyze online learning activity data in R using a histogram, boxplot, bubble scatter plot and time-series line chart, and to build an interactive Tableau dashboard.

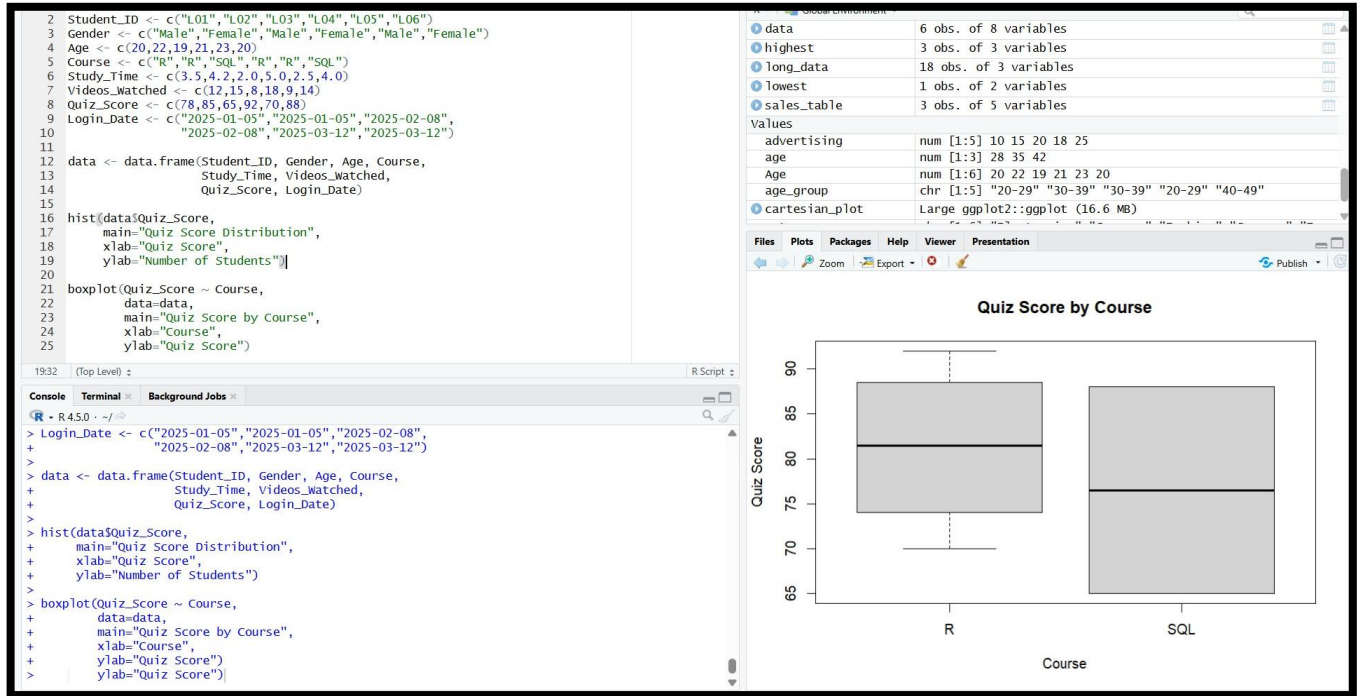
Dataset: Online_Learning_Activity.csv

Student ID	Gender	Age	Course	Study Time (hrs)	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Question 1: Histogram and Boxplot of Quiz Scores

Question: Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

Aim: To study the distribution of quiz scores and compare the two courses using a histogram and a boxplot.

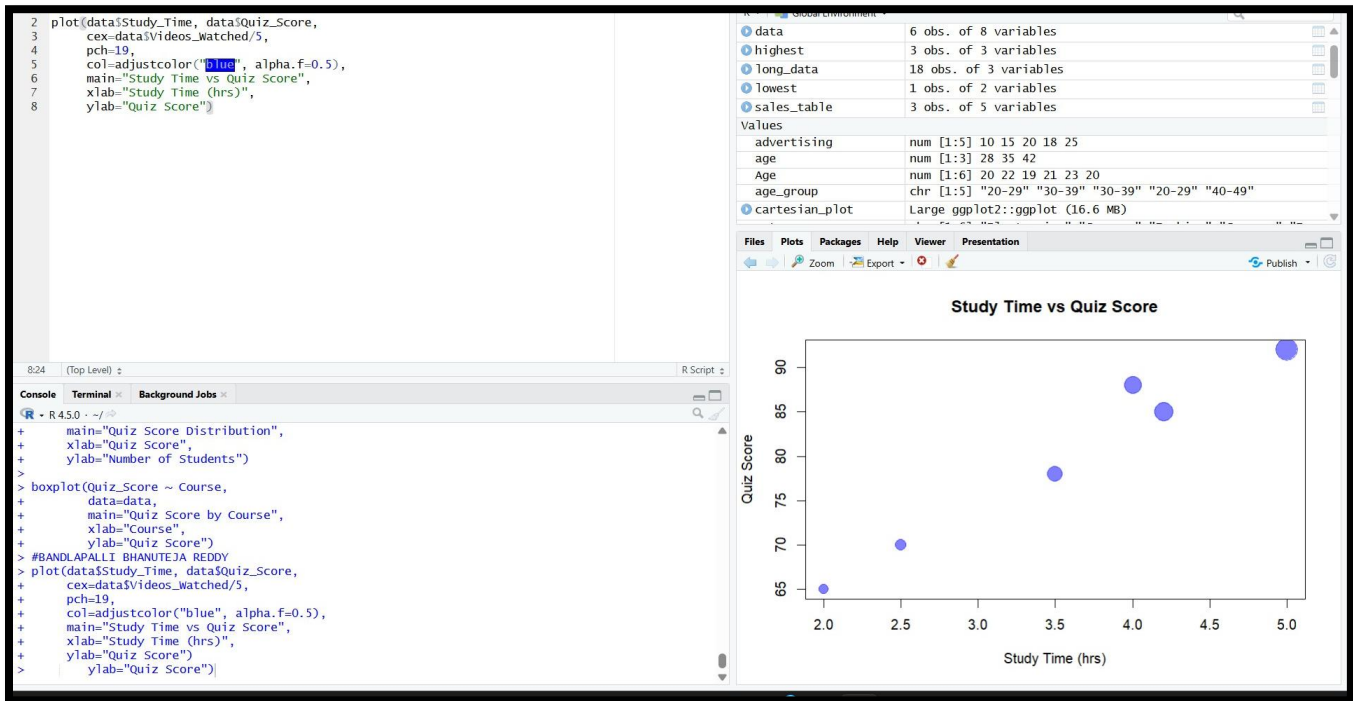


Result: Scores range from 65 to 92 with a mean of about 79.7. R students average 81.25 and SQL students 76.5, so R students perform slightly better, and both courses have a similar spread.

Question 2: Bubble Scatter Plot of Study Time vs Quiz Score

Question: Create a scatter plot using R programming between `Study_Time` and `Quiz_Score`. Represent `Videos_Watched` using bubble size. Apply transparency. Explain the learning pattern.

Aim: To explore how study time and videos watched relate to quiz scores using a transparent bubble scatter plot.

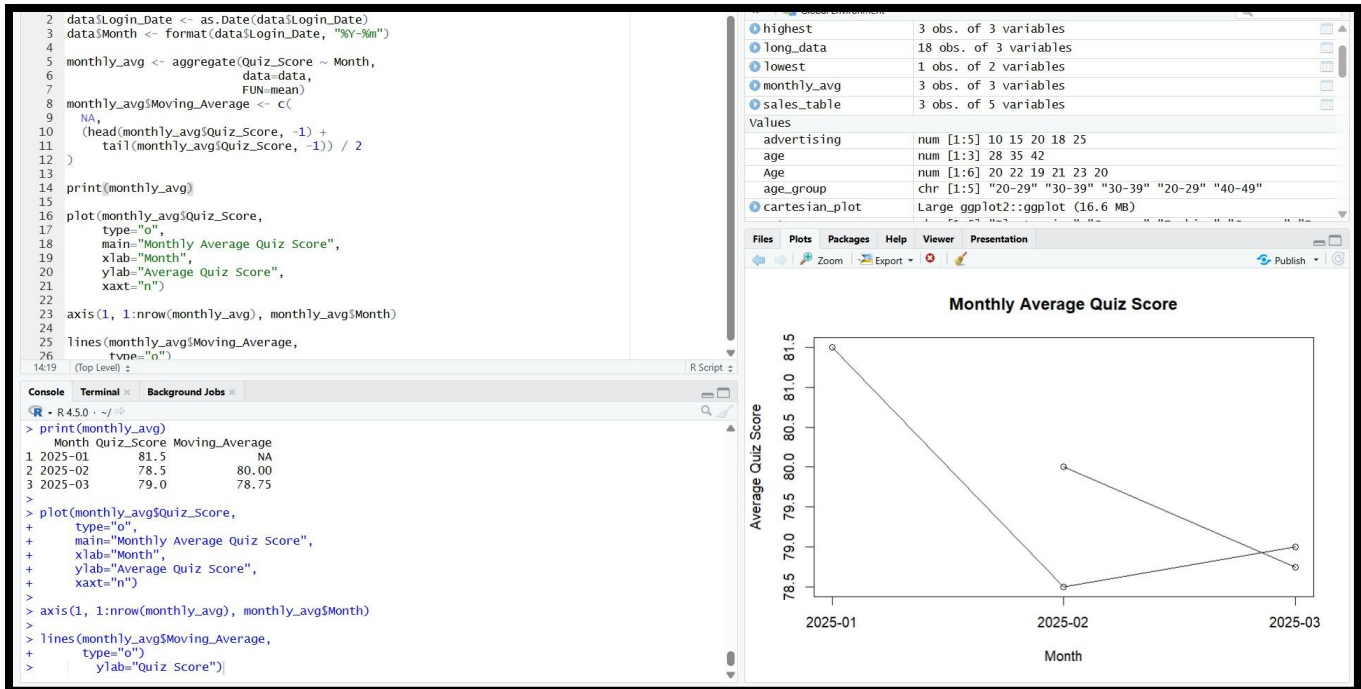


Result: Students who study more score higher (correlation about 0.98, roughly 9 more marks per extra hour), and those who watch more videos have bigger bubbles and higher scores, so learning effort pays off.

Question 3: Monthly Average Quiz Score Trend

Question: Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

Aim: To convert Login_Date to a date format and show the monthly average quiz score with a smoothed trend line.

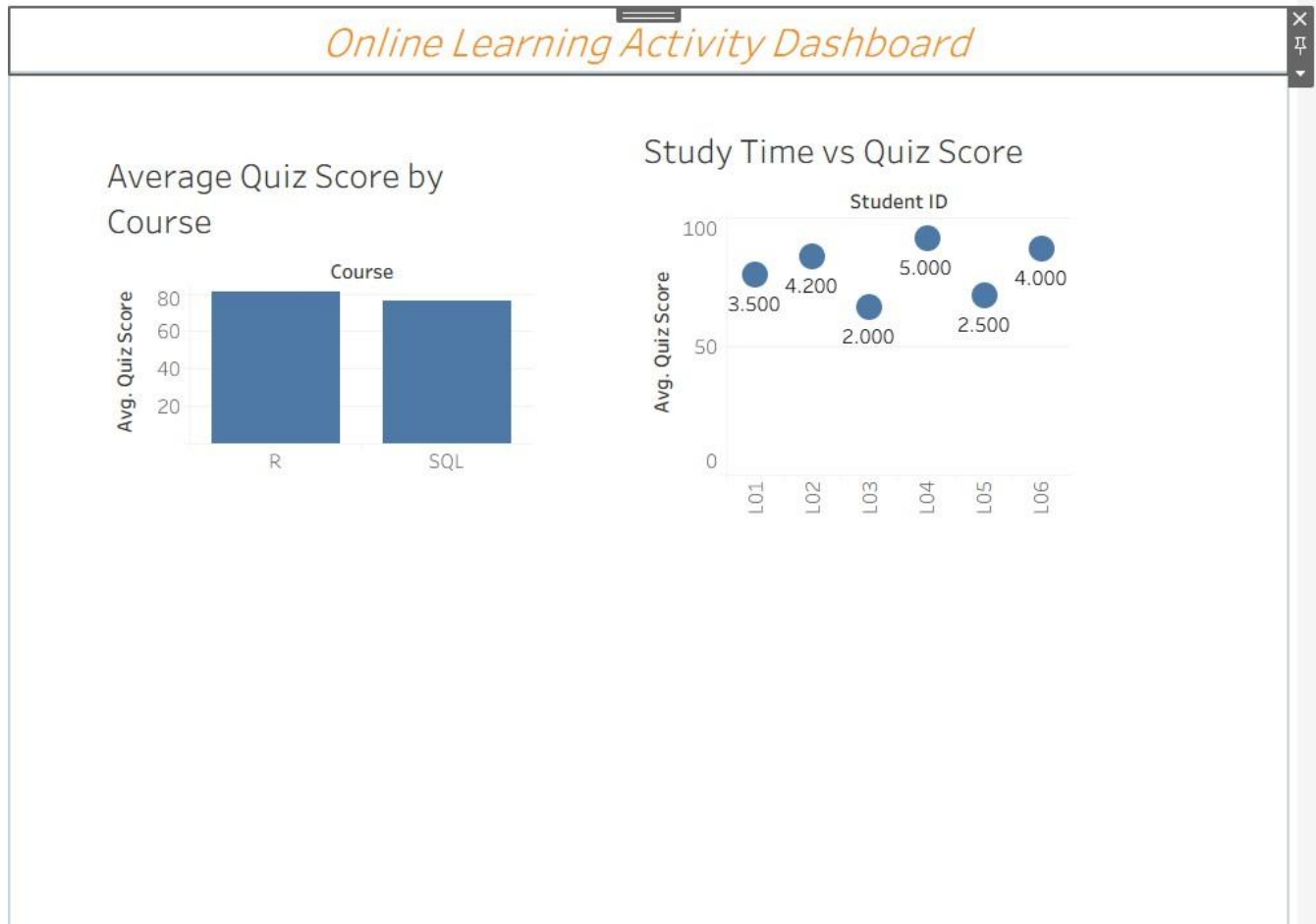


Result: Monthly averages are 81.5 in January, 78.5 in February and 79.0 in March, so scores dip slightly after January and then stay steady around 79.

Question 4: Interactive Tableau Dashboard

Question: Import `Online_Learning_Activity.csv` into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

Aim: To build a mini interactive Tableau dashboard with a bar chart, a scatter plot and filters for Gender and Course.



Result: The bar chart shows R at 81.25 and SQL at 76.5, the scatter plot shows scores rising with study time, and the Gender and Course filters update both views; female students average 88.3 and male students 71.0.

Overall Result

Observation: The average study time is about 3.5 hours and the average quiz score is about 79.7, and students who study longer score clearly higher.